



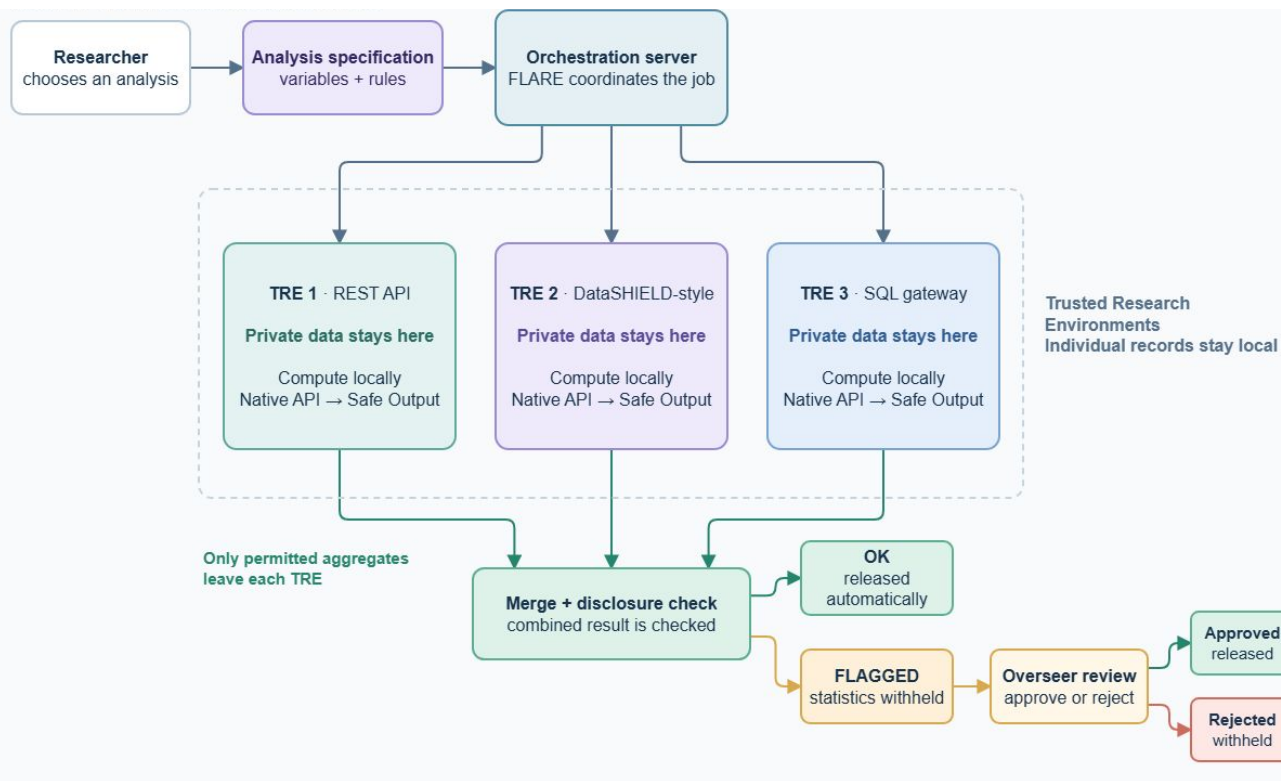
HEIMDALL

Team:

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Federated analysis across TREs with different APIs

Team 1 · Nordic Biobank X NVIDIA 2026 hackathon



Request

type, canonical variables, filters, min_cell_size, project_id.



Dispatch

The FLARE controller broadcasts the spec and collects results



Local execution

Executor loads the adapter for its own tre_id, calls the native API.



Safe output

The TRE filters results before passing them to the FLARE client



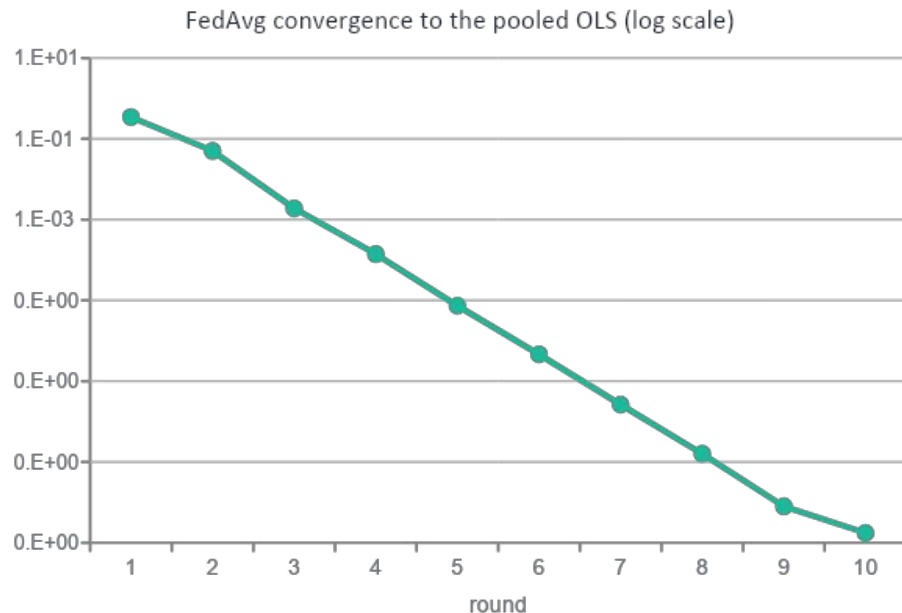
Aggregation and review

The server combines results and checks disclosure risk.

Use case - Linear regression

$$y \sim \text{age} + \text{sex} + \text{bmi} + \text{ldl} + \text{rs001} + \text{rs007} + \text{rs013}$$

coefficient	exact (1 round)	FedAvg (10 rounds)	pooled OLS
intercept	89.2729	89.2729	89.2729
age	0.5055	0.5055	0.5055
sex	4.2034	4.2034	4.2034
bmi	0.8062	0.8062	0.8062
ldl	1.2189	1.2189	1.2189
snp_rs001	2.3934	2.3934	2.3934
snp_rs007	-1.2817	-1.2817	-1.2817
snp_rs013	0.9003	0.9003	0.9003
max error	1.7e-11	1.8e-11	—



Testing and deployment status: 3 APIs

Cloud demo, TRE demo on-device, UI proof of concept



In-process

Unit tests:
adapters, filter,
merge, disclosure
check, overseer



FLARE simulator

3 to 100 simulated
TREs, with exact
results at each
tested size



Real FLARE processes

Server and 3 clients
using mTLS gRPC and
admin-API submission.
With one client
stopped, 2 sites
reported.



Docker

7 images on isolated
networks.
Jobs run inside
flare-server.
With one container
stopped, 2 sites
reported.

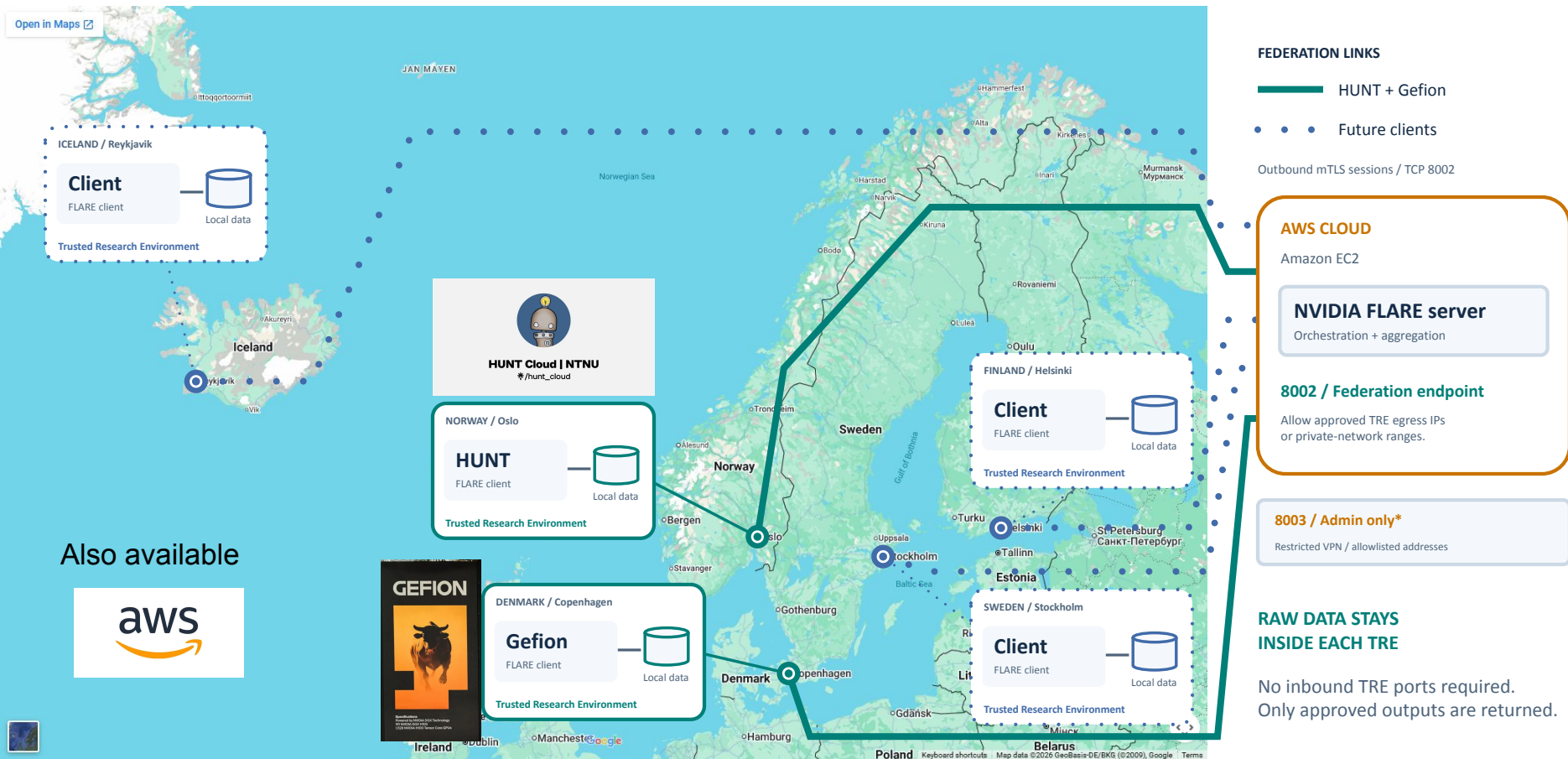


next

Cloud

Clients on HUNT
Cloud, Gefion and
NextCloud, server on
Brev/AWS.
Each TRE needs an
outbound firewall
rule.

Nordic TREs, server and admin on AWS



```

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS      PORTS      NAMES
ubuntu@ip-172-31-26-165:~/workspace/example_project/prod_00/server1$ docker run --rm -it --name nvflare-server --user "$(id -u):$(id -g)" -p 8002:8002 -p 8002:8002 -v "$PWD":/workspace -v "$PWD/runtime":/tmp/nvflare -w /workspace nvflare:2.9.0 bash startup/sub_start.sh
WORKSPACE set to /workspace
PYTHONPATH is /local/custom:
start fl because process of 16 does not exist
new pid 17
2026-09-17 15:20:02,142 - ServerDeployer - INFO - server heartbeat timeout set to 600
2026-09-17 15:20:02,240 - CoreCell - INFO - server: creating listener on http://0:8002?admin_listener=true
2026-09-17 15:20:02,580 - CoreCell - INFO - server: created backbone external listener for http://0:8002?admin_listener=true
2026-09-17 15:20:02,581 - conn_manager - INFO - Connector [CH00002 PASSIVE tcp://127.0.0.1:36551] is starting
2026-09-17 15:20:03,083 - CoreCell - INFO - server: created backbone internal listener for tcp://localhost:36551
2026-09-17 15:20:03,083 - conn_manager - INFO - Connector [CH00001 PASSIVE http://0:8002?admin_listener=true] is starting
2026-09-17 15:20:03,085 - Cell - INFO - Register blob CB for channel='aux_communication', topic='*'
2026-09-17 15:20:03,086 - FederatedServer - INFO - max_reg_duration=60.0
2026-09-17 15:20:03,087 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Request
2026-09-17 15:20:03,088 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Abort
2026-09-17 15:20:03,089 - AuxRunner - INFO - registered aux handler for topic fed.event
2026-09-17 15:20:03,091 - ServerDeployer - INFO - deployed FLARE Server.
2026-09-17 15:20:03,097 - Cell - INFO - Register blob CB for channel='hci_channel', topic='*'
2026-09-17 15:20:03,098 - hci - INFO - Admin Server is started
2026-09-17 15:20:03,099 - root - INFO - Server started
2026-09-17 15:26:46,170 - conn_manager - INFO - Connection [CN00002 172.17.0.2:8002 <= 172.31.16.187:38904 SSL site-2] is created: PID: 17
2026-09-17 15:26:46,368 - ClientManager - INFO - Client: New client site-2@172.17.0.2 joined. Sent token. Total clients: 1
2026-09-17 15:26:46,369 - FederatedServer - INFO - client site-2 registered with site_config keys: ['allow_log_streaming']

```

SERVER (AWS)

CLIENT (HUNT Cloud)

```

(team1) ubuntu@ncfh26-hackathon-iaas-aLula:~/users/espen.hagen/site-1/startup$ grep -q "server1" /etc/hosts || echo "16.170.143.132 server1"
; ./sub_start.sh --once
16.170.143.132 server1
WORKSPACE set to /home/ubuntu/users/espen.hagen/site-1
PYTHONPATH is /local/custom:
Start NVFlare directly
2026-09-18 12:41:19,575 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Request
2026-09-18 12:41:19,576 - AuxRunner - INFO - registered aux handler for topic ObjectStreamer.Abort
2026-09-18 12:41:19,745 - CoreCell - INFO - site-1: created backbone external connector to http://server1:8002
2026-09-18 12:41:19,746 - conn_manager - INFO - Connector [CH00002 PASSIVE tcp://127.0.0.1:58822] is starting
2026-09-18 12:41:20,247 - CoreCell - INFO - site-1: created backbone internal listener for tcp://localhost:58822
2026-09-18 12:41:20,248 - conn_manager - INFO - Connector [CH00001 ACTIVE http://server1:8002] is starting
2026-09-18 12:41:20,250 - Cell - INFO - Register blob CB for channel='server_command', topic='get_task'
2026-09-18 12:41:20,251 - Cell - INFO - Register blob CB for channel='server_command', topic='submit_update'
2026-09-18 12:41:20,251 - FederatedClient - INFO - Wait for engine to be created.
2026-09-18 12:41:20,333 - conn_manager - INFO - Connection [CN00002 10.5.5.115:46984 => 16.170.143.132:8002 SSL server1]
2026-09-18 12:41:20,576 - Cell - INFO - Register blob CB for channel='aux_communication', topic='*'
2026-09-18 12:41:20,578 - AuxRunner - INFO - registered aux handler for topic fed.event
2026-09-18 12:41:20,685 - Authenticator - INFO - verified server identity 'server1'
2026-09-18 12:41:20,804 - Authenticator - INFO - Verified received token and signature successfully
2026-09-18 12:41:20,805 - FederatedClient - INFO - Successfully registered client:site-1 for project example_project.
2026-09-18 12:41:20,807 - FederatedClient - INFO - Got engine after 0.5561730861663818 seconds
2026-09-18 12:41:20,808 - FederatedClient - INFO - Connected to server: http://server1:8002

```

Swagger - User endpoint for aggregated results

127.0.0.1:8500/docs#/

Bifrost federated analysis API 0.2.0 OAS 3.1

[/openapi.json](#)

Bifrost HTTP API -- the contract the researcher UI (frontend) talks to.

```
SERVER_OUT=server/api_out uvicorn server.api:app --port 8500 --workers 1
```

Start mock TREs separately with `python scripts/dev_tres.py`. Neither TREs nor FLARE are required for startup or /health. API_TRE_TIMEOUT (seconds per HTTP operation, default 10) and API_TRE_WORKERS (default 8) control adapter fan-out. API_CORS_ORIGINS (comma-separated, default http://localhost:3000) allows the UI origin. Use an exclusive SERVER_OUT directory; see analysis_service's concurrency and demo-versus-secure-deployment notes. Variable names are canonical (snp_rs001), never local column names; requests carry variable names, never participant data. Response shapes: server/schemas.py, exported to docs/schemas/ and served at /openapi.json.

default

GET	/health	Health	▼
GET	/allele-frequency	Allele Frequency	▼
POST	/linear-regression	Linear Regression	▼
POST	/logistic-regression	Logistic Regression	▼
GET	/metadata	Metadata	📄 ▼
GET	/examples/{name}	Example	▼
POST	/run	Submit Run	▼
GET	/run/{run_id}	Run Status	▼
GET	/overseer	Overseer	▼
POST	/overseer/{spec_hash}/{decision}	Overseer Decide	▼
GET	/audit	Audit	▼

Schemas

AlleleCounts > Expand all object

Use case 1 - calculate allele freq

GET /allele-frequency Allele Frequency

Parameters

Name	Description
variant * required	
string (query)	snp_rs001
	minLength: 1
project_id * required	
string (query)	ncfh-2026-demo
	minLength: 1
min_cell_size	
integer (query)	5
	minimum: 5

Input

- Variant
- Project id

```
{
  "rejected_per_site": {},
  "stats": {
    "snp_rs001": {
      "genotype_counts": {
        "0": 25690,
        "1": 4143,
        "2": 157
      },
      "allele_counts": {
        "minor": 4477,
        "major": 55523,
        "n_alleles": 60000
      },
      "allele_freq": 0.07461666666666666,
      "allele_freq_sites": [
        "brev",
        "gefion",
        "hunt"
      ]
    }
  }
}
```

Output (json)

- Allele frequency

Use case 2 - linear regression (SBP ~ age + BMI)

POST /linear-regression Linear Regression

Parameters

No parameters

Request body * required

Edit Value | Schema

```
{
  "features": ["age", "bmi"],
  "target": "sbp",
  "project_id": "ncfh-2026-demo",
  "filters": {},
  "min_cell_size": 5
}
```

Input (json)

- Outcome + predictor variables

Output (json)

- Regression coefficients

```
{
  "region_per_site": {
    "hunt": "nordic",
    "gefion": "nordic",
    "brev": "nordic"
  },
  "rejected_per_site": {},
  "stats": {
    "_linreg": {
      "ols": {
        "outcome": "sbp",
        "coef": {
          "intercept": 95.52337125694957,
          "age": 0.5058367368789758,
          "bmi": 0.8092701406242438
        }
      },
      "n": 30000
    }
  }
}
```

Preliminary User Experience: What we suggest for the future

The screenshot displays the BIFROST Federated analysis gateway interface. The browser address bar shows `http://localhost:3000`. The header bar is dark blue with the BIFROST logo and text "BIFROST Federated analysis gateway". On the right of the header, there is a "Secure session" button and a user profile icon labeled "RW".

On the left, a "RESEARCHER MENU" is visible with the following items:

- Analysis (highlighted)
- Overseer queue
- Audit logs

The main content area is titled "BIFROST / ANALYSIS WORKSPACE" and features a large heading "Cross-TRE analysis". Below this heading is a subtext: "Build a request, run it across connected environments, and inspect disclosure-safe output." A button labeled "Copy result JSON" is located to the right of the subtext.

A yellow informational banner states: "Record-level data stays inside each TRE. Only aggregate statistics and model information pass through the gateway."

Below the banner, two analysis options are presented:

- Allele frequency** (Federated analysis) - This option is highlighted with a blue border.
- Linear regression** (Federated learning)

The interface is divided into two main steps:

STEP 1: Define your analysis

This step includes a "Safe project" dropdown menu currently set to "ncfh-2026-demo". Below this, there is a section for "Genotype variables" with a list of checkboxes:

- ☒ snp_rs001
- ☐ snp_rs002
- ☐ snp_rs003
- ☐ snp_rs004
- ☐ snp_rs005
- ☐ snp_rs006

A "Load example..." button is located at the top right of this section.

STEP 2: Connected TREs

This step shows a list of connected TREs, with a status indicator "3 online" at the top right. The list includes:

- hunt** (rest - NO) with a status of "Ready".
- gefion** (datashield - NO) with a status of "Ready".

Next steps



Cloud deployment

Configure the server-client setup with real TREs



Real variable names and variants

Replace the invented local column names and rs001–rs020 with each site's schema and real rsIDs



Researcher interface

Build a page for selecting variables and running analyses, with results, coverage and audit records.



Hierarchical aggregation

Add regional relay aggregators through FLARE's hierarchical topology, using the existing associative merge.

Thank You!

